中国科学：生命科学

•致编辑的信•

2023年9月 第66卷 第9期：2189–2192
https://doi.org/10.1007/s11427-022-2318-4

TCMSTD 1.0：传统中医药系统毒理数据库的系统分析

宋莉莉[1†]，钱文秀[1†]，尹宏庆[1]，，孙宇[2]，，孙秀燕[1]，，李国辉[1]，，何俊杰[1]，，郑燕雪[1]，，张悦[1]，，王佳怡[1] & 李昱波[1*]

[1]中国天津市天津中医药大学, 邮编*301617;*[2]中国天津市天津中医药大学宝康医院, 邮编 *300073*

收到日期：2022年12月19日；接受日期：2023年3月7日；网络发表日期：2023年4月4日

引用：宋磊，钱伟，尹浩，孙阳，孙翔，李刚，何建，郑燕，张颖，王杰，等。（2023）。TCMSTD1.0：中医药系统毒理学数据库的系统分析。《中国科学：生命科学》66，2189–2192。https://doi.org/10.1007/s11427-022-2318-4

尊敬的编辑，

传统中医药（TCM）的安全性和现代化是关系到用户生命和健康的重要问题。近年来，由中医药和传统无毒中医药引起的不良反应频繁被报道（Joo 等，2020；Liu 等，2021）。随着中医药安全性持续受到广泛关注，许多研究者一直在研究中医药的毒性，涉及毒性成分、作用机制以及中医药的解毒策略（支持信息中的图S1）。尽管一些中医药会引起毒性反应，但它们在治疗某些不可忽视的疾病中也发挥着关键作用（Fang 等，2022）。因此，毒性研究对于中医药的开发和安全使用至关重要且迫切需要。

目前，数据库技术能高效且方便地帮助研究人员挖掘、存储、检索和分析中药信息。利用数据库技术系统地整合和分析中药研究成果，将促进并有助于中药临床研究（Luo 等, 2022）。然而，中药及其成分的药理作用是现有中药数据库的主要关注点（Fang 等, 2021；Ru 等, 2014；Xu 等, 2019）。此外，现有数据库并未区分有毒和药理作用的成分。对靶标的动态作用，对特定毒性表现的靶标进行分类和总结，并支持以往研究的证据（Davis 等，2021；Lim 等，2010）。考虑到上述问题，我们构建了中药系统毒理学数据库（TCMSTD；https://www.bic.ac.cn/TCMSTD/），这是首个在国内外专注于中药安全使用的数据库。本研究旨在系统整合和分析有毒中药的研究成果，并提供中药成分毒性作用及靶标预测功能，以促进中药毒性研究。

TCMSTD 的构建分为两个主要阶段。第一阶段涉及数据收集和处理（图 1A）。我们尝试使用计算机技术提取文献内容。然而，我们确定，由于文献格式和内容书写方式多样，计算机信息提取方法会导致信息遗漏和提取错误。因此，我们采用了人工数据收集方法。TCMSTD 的纳入标准和最终确定模块（支持信息中的表 S1）如下：（i）有毒中药（252 种）：我们从 10 部权威中药著作（支持信息中的表 S2）总结出 9,139 种中药，并筛选出具有临床中毒报告或通过毒理实验验证毒性的中药，最终纳入 252 种有毒中药

† 对本工作贡献相同 *通讯作者（邮箱：yaowufenxi001@sina.com）

SCIENCE CHINA
Life Sciences

•LETTER TO THE EDITOR•

September 2023 Vol.66 No.9: 2189–2192
https://doi.org/10.1007/s11427-022-2318-4

TCMSTD 1.0: a systematic analysis of the traditional Chinese medicine system toxicology database

Lili Song[1†], Wenxiu Qian[1†], Hongqing Yin[1], Yu Sun[2], Xiuyan Sun[1], Guohui Li[1], Junjie He[1], Yanxue Zheng[1], Yue Zhang[1], Jiayi Wang[1] & Yubo Li[1*]

[1]*Tianjin University of Traditional Chinese Medicine, Tianjin 301617, China;*
[2]*Baokang Hospital of Tianjin University of Traditional Chinese Medicine, Tianjin 300073, China*

Received December 19, 2022; accepted March 7, 2023; published online April 4, 2023

Citation: Song, L., Qian, W., Yin, H., Sun, Y., Sun, X., Li, G., He, J., Zheng, Y., Zhang, Y., Wang, J., et al. (2023). TCMSTD 1.0: a systematic analysis of the traditional Chinese medicine system toxicology database. Sci China Life Sci 66, 2189–2192. https://doi.org/10.1007/s11427-022-2318-4

Dear Editor,

The safety and modernization of traditional Chinese medicine (TCM) are significant concerns for users' lives and health. Recently, the adverse effects caused by TCM and traditional nontoxic TCM have been frequently reported (Joo et al., 2020; Liu et al., 2021). As the safety of TCM continues to attract widespread attention, many researchers have been investigating the toxicity of TCM in regard to the toxic ingredients, effects, mechanisms, and detoxification strategies in TCM (Figure S1 in Supporting Information). Although some TCMs cause toxic reactions, they also play key roles in treating certain unignorable diseases (Fang et al., 2022). Therefore, toxicity research is crucial and urgently needed to develop and safely use TCM.

Currently, database technology efficiently and conveniently aids researchers in mining, storing, searching, and analyzing the information on TCMs. The use of database technology to systematically integrate and analyze the research results of TCM will promote and help clinical research on TCM (Luo et al., 2022). However, the pharmacological effects of TCMs and their constituents are the main emphases of existing TCM databases (Fang et al., 2021; Ru et al., 2014; Xu et al., 2019). Moreover, the current databases do not distinguish between toxic and pharmacodynamic effects on targets, classify and summarize targets for specific toxic manifestations, and support evidence from previous studies (Davis et al., 2021; Lim et al., 2010). Considering the aforementioned issues, we constructed the Traditional Chinese Medicine System Toxicology Database (TCMSTD; https://www.bic.ac.cn/TCMSTD/), which is the first database locally and internationally that focuses on the safe use of TCMs. This study aimed to systematically integrate and analyze the research results of toxic TCMs and provide the functions of the toxic effect and toxic target prediction of TCM ingredients to enable TCM toxicity research.

The construction of TCMSTD was divided into two main stages. The first stage involved data collection and processing (Figure 1A). We attempted to extract the literature content using computer technology. However, we determined that, because of the varied literature format and content writing manner, the computer information extraction method will cause information omissions and extraction errors. Thus, we used the manual data collection method. The inclusion criteria and finalization modules for TCMSTD (Table S1 in Supporting Information) are as follows: (i) toxic TCMs (252 species): we summarized 9,139 TCMs from 10 authoritative works on TCMs (Table S2 in Supporting Information) and screened out the TCMs with clinical poisoning reports or proven toxicity through toxicological experiments, consequently including 252 kinds of toxic

†Contributed equally to this work
*Corresponding author (email: yaowufenxi001@sina.com)

数据库中的中药。为了精确识别有毒中药，我们从天津中医药大学中医药资源教学与科研科及保康医院中药房收集了150种中药，并根据中药的特征对其进行了拍照。（ii）方剂（22种）：采集注射剂和消毒剂以外的方剂，依据国家药品监督管理局发布的不良药物反应信息通报。（iii）化学成分（4,361种）：中药成分主要基于过去五年发表的综述文章收集，并尽可能排除挥发性成分。（iv）有毒成分（226种）：除了符合项目(iii)的标准外，仅被毒理实验证实的成分才被认为是有毒成分。（v）毒性靶点（2,425种）：我们对其进行了系统分类

数据库构建的第二阶段涉及根据第一阶段收集的数据设计和构建数据库的模块、功能和接口（图 1B）。TCMSTD 作为一个 Web 应用程序实现，前端开发使用 JavaScript 和 HTML。使用的核心 JavaScript 库包括 Vue.js（https://vuejs.org/）作为主要前端框架，plotly.js（https://plotly.com/）用于制作棒棒糖图，以及 vis.js（https://visjs.org）用于网络展示。高层 Web 框架 Django（https://www.djangoproject.com/）用于后端数据预处理和分析。开源数据管理系统 MySQL 用于表数据的保存和访问。

TCMSTD 具有中文和英文界面，用于设置五个功能（支持信息中的图 S4）。这五个功能如下：（i）浏览功能：TCMSTD 具有中文和英文的浏览界面，用户可以点击数据库首页的“浏览”按钮，或者直接点击底部的目标模块

首页。(ii) 搜索功能：全局搜索功能基于 Elasticsearch 模块。TCMSTD 提供五种元素的搜索选项（即中药、方剂、化学成分、毒性成分和毒性靶点），用户可以在数据库首页右上角的搜索框中看到相应的关键词输入，并进入详情页，从而实现该功能。(iii) 预测功能：TCMSTD 内嵌两个子数据库，即毒性成分数据库和毒性靶点数据库。它们可以通过预测成分是否有毒以及预测毒性成分的毒性靶点来实现两种功能。该功能使用基于 Open Babel 包的 Tanimoto 系数来评估化学成分的相似性。默认情况下，查询化学物质在 TCMSTD 中有 Tanimoto 系数大于 0.6 且至少存在一个分子时，将被视为具有相似的毒性靶点。用户可以点击“工具” o

数据更新是数据库一项极其重要的任务，其目的是确保及时显示当前中医药的研究进展，并体现数据库存在的意义。中医药标准数据库（TCMSTD）每六个月更新一次，每年6月30日和12月30日发布新版本。我们始终关注有毒中药的研究进展，以便及时更新数据库中无效的数据，并补充缺失的数据和最新的研究成果。在随后的更新版本中，我们将根据计算机的发展和中药毒性研究人员的实际工作需要，更新和升级算法及程序。作为开发团队，我们拥有数据库后台的管理权限，因此会密切关注数据库的状态。

TCMs in the database. To elaborately identify toxic TCMs, we collected 150 TCMs from the Teaching and Research Section of Traditional Chinese Medicine Resources of Tianjin University of Traditional Chinese Medicine and Pharmacy of Baokang Hospital and photographed them according to the characteristics of TCM. (ii) Formulas (22 species): formulas excluding those of injections and sanitizers were collected according to the Adverse Drug Reaction Information Bulletin issued by the National Medical Products Administration (China). (iii) Chemical ingredients (4,361 species): TCM ingredients were primarily collected based on review articles published over the last five years, and volatile ingredients were excluded as much as possible. (iv) Toxic ingredients (226 species): in addition to meeting the standards of Item (iii), only the ingredients confirmed by toxicological experiments will be considered toxic ingredients. (v) Toxic targets (2,425 species): we systematically classified and summarized five toxic targets according to the most significant toxicity manifestations caused by TCMs, namely, cardiotoxicity, hepatotoxicity, nephrotoxicity, neurotoxicity, and reproductive toxicity (Figure S2A in Supporting Information), constituting the toxic target sub-database of TCMSTD. The aforementioned contents refer to databases, including TCMSP (Ru et al., 2014), PubChem (Kim et al., 2021), ChemSpider (Pence and Williams, 2010), NCBI (Sayers et al., 2021), UniProt (UniProt, 2021), OMIM (Hamosh et al., 2021), and DisGeNET (Piñero et al., 2021). The data are mainly obtained from five literature websites, including CNKI, VIP, WanFang, PubMed, and Web of Science. A total of 5,522 valid articles were included. The aforementioned five modules are interconnected (Figure S2B in Supporting Information). Figure S3 in Supporting Information shows the specific data types that are incorporated in TCMSTD.

The second stage of database construction involved designing and building the modules, functions, and interfaces of the database according to the data collected in the first stage (Figure 1B). TCMSTD was implemented as a web application using JavaScript and HTML for front-end development. The core JavaScript libraries used include Vue.js (https://vuejs.org/) as the main front-end framework, plotly.js (https://plotly.com/) for Lollipop chart creation, and vis.js (https://visjs.org) for network display. The high-level web framework Django (https://www.djangoproject.com/) was used for back-end data preprocessing and analysis. The open-source data management system MySQL was used for table data saving and accessing.

The TCMSTD has Chinese and English interfaces to set up five functions (Figure S4 in Supporting Information). These five functions are as follows: (i) browse function: TCMSTD has browsing interfaces in Chinese and English, and the user can click the "Browse" button on the home page of the database or directly click the target module at the bottom of the home page. (ii) Search function: the global search function is based on the Elasticsearch module. TCMSTD provides search options for five elements (i.e., TCM, formula, chemical ingredient, toxic ingredient, and toxic target), and the user can see the corresponding keyword input in the search box at the upper right corner of the home page of the database and proceed to the details page, thus realizing the function. (iii) Predictive function: two sub-databases are embedded in TCMSTD, namely, the toxic ingredient and toxic target databases. They can realize two functions by predicting whether the ingredient is toxic or not and predicting the toxic target of a toxic ingredient. This function uses the Tanimoto coefficient based on the Open Babel package to evaluate the similarity of chemical ingredients. By default, query chemicals that have a Tanimoto coefficient greater than 0.6 and one molecule in TCMSTD would be treated as having similar toxic targets. Users can click the "Tools" option on the home page and enter the ingredient structure in the blank space to use this function. The results page will show the list of results and network diagram. Table S3 in Supporting Information shows the list of toxic target results of predicted toxic ingredients. (iv) Systematic analysis function: the "System Analysis Function" option is displayed on the home page of the database, and it is available upon click. The required elements (i.e., TCM, formula, ingredient, and toxic target) are selected, and their network relationships are built based on the network module of the visualization database vis.js (https://visjs.org). Pathway enrichment analysis of the Gene Ontology and Reactome pathway information is conducted. The enriched pathway is computed using a hypergeometric test to derive the significant p-value, which is further corrected using the Benjamini and Hochberg algorithm. (v) Download function: the TCMSTD can download data tables and predictive analysis results. Each part of the downloadable content is provided with text or icon prompts. Even without user registration or login, the aforementioned five features can be enabled quickly, conveniently, and freely. Figure S5 in Supporting Information shows the application case of TCMSTD.

Data update is an extremely important task of the database, which aims to ensure a timely display of the current research progress on TCM and reflect the significance of the existence of the database. Every six months, TCMSTD will be updated, and new versions will be released on June 30 and December 30 yearly. We always pay attention to the research progress on toxic TCMs to promptly update the invalid data in the database and supplement the missing data and the latest research results. In subsequent updated versions, we will update and upgrade the algorithm and program according to the development of computers and the actual work needs of TCM toxicity researchers. As a development team, we have administrative rights to the database background; thus, close attention will be paid to the database status at any

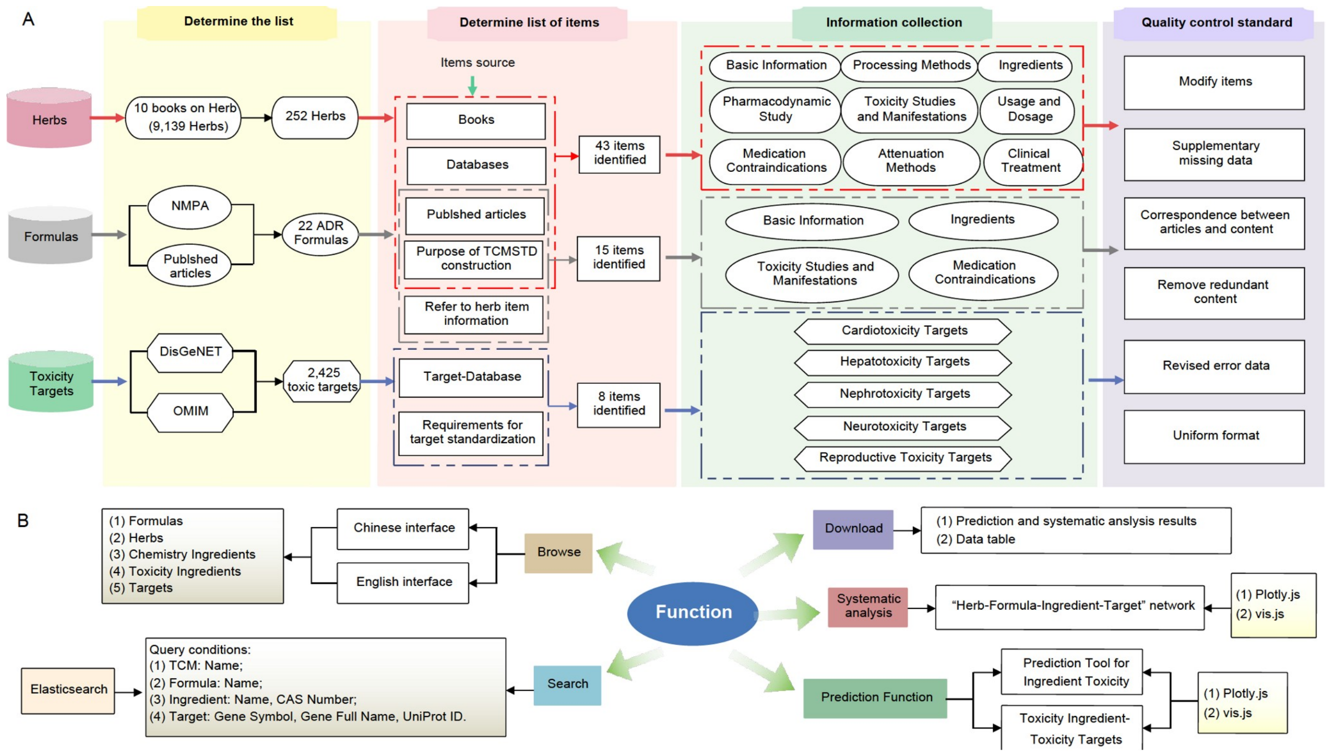

图1 数据库构建过程：（A）数据准备过程；（B）中药系统毒理学数据库的功能构建。

是时候识别问题并做出适当响应，确保用户无问题地使用数据库。

总之，TCMSTD 是首个专注于中医药安全性的数据库，具有三大创新：第一，挖掘并进行中药毒性信息的系统整合与分析；第二，在 TCMSTD 中嵌入涵盖五种毒性作用的毒性靶点子数据库；第三，利用 TCMSTD 基于成分的结构信息预测毒性。一个合格的数据库应具备合理的数据结构、较小的数据冗余度，以及信息和分析系统中包含的准确可靠的结果，并能够实时追踪中药毒性研究状态，以便进行及时的整合与分析，这也是 TCMSTD 建设的目的。我们有信心，TCMSTD 将成为促进中医药安全使用的有效工具，并为从事中药毒性研究的科研人员、中药安全监管人员以及药物使用者提供一个高效且全面的中药毒理信息共享平台。

合规与道德 作者声明他们没有利益冲突。

致谢 本研究得到中国国家自然科学基金（*81873194*、*81573825*、*81273998*）和"青年启晃学者"国家人才项目（*2020*）的资助。作者感谢*EHBIO*基因科技（北京）有限公司的薛濮先生和刘墨宇先生在*TCMSTD*构建过程中提供的帮助。

参考文献

Davis, A.P., Grondin, C.J., Johnson, R.J., Sciaky, D., Wiegers, J., Wiegers, T.C., 和 Mattingly, C.J. (2021)。比较毒理基因组数据库（CTD）：2021 年更新。核酸研究 49, D1138–D1143。Fang, S.S., Dong, L., Liu, L., Guo, J.C., Zhao, L.H., Zhang, J.Y., Bu, D.C., Liu, X.K., Huo, P.P., Cao,W.C., 等. (2021)。HERB：一个高通量实验和参考指导的中药数据库。核酸研究 49, D1197–D1206。Fang, Z., Guo, Y., Wang, Z., He, T., Wang, J., Bai, Z., 和 Xiao, X. (2022)。细辛消耗与患者肝细胞癌无关：一项回顾性队列研究。中草药 14, 470–475。Hamosh, A., Amberger, J.S., Bocchini, C., Scott, A.F., 和 Rasmussen, S. A. (2021)。在线孟德尔遗传学数据库（OMIM®）：Victor McKusick 的杰作。美国医学遗传学杂志 A 185, 3259–3265。Joo, Y.J., Newcombe, D., Nosa, V., 和 Bullen, C. (2020)。研究槟榔使用、前因和后果：一项回顾性研究

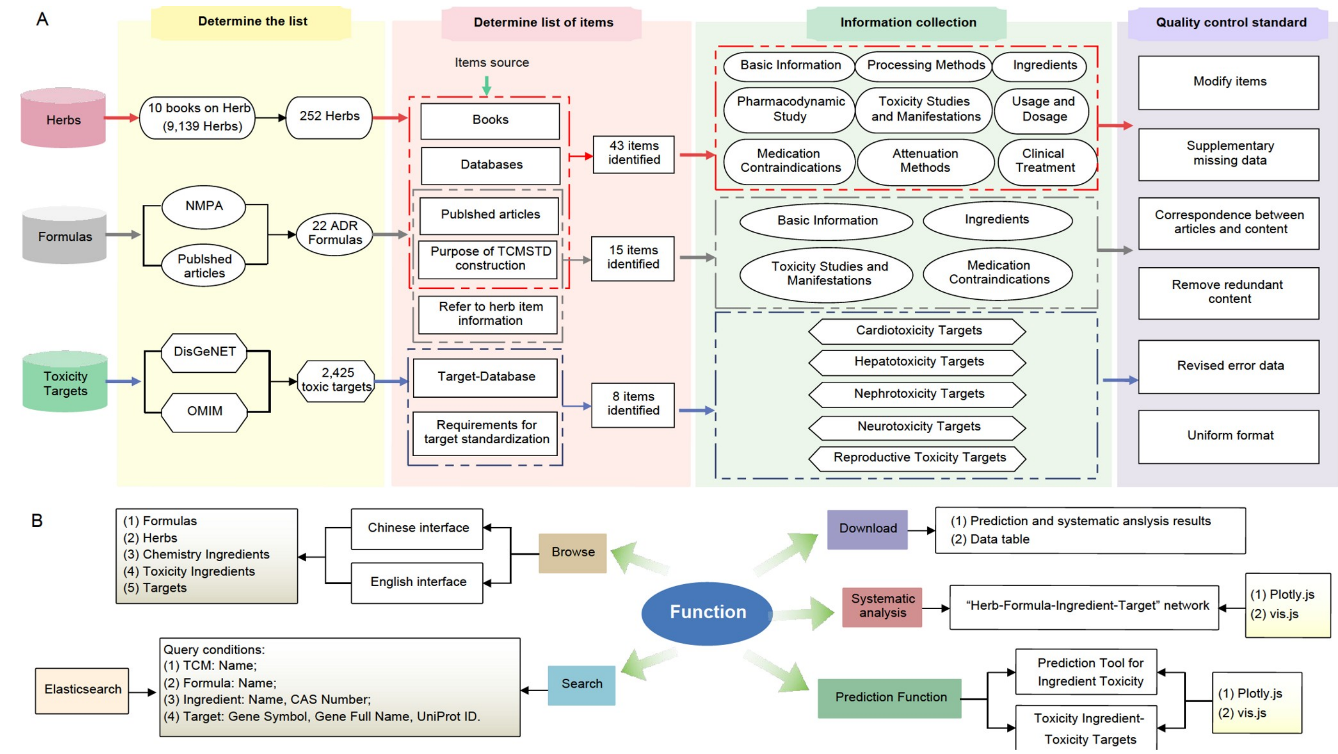

Figure 1 Database construction process: (A) data preparation process and (B) functional construction of Traditional Chinese Medicine System Toxicology Database.

time to identify problems and respond appropriately, ensuring problem-free utilization of the database by users.

In summary, TCMSTD, the first database focusing on TCM safety, has three innovations: first, mining and system integration and analysis of toxic-TCM information; second, the toxic target sub-database covering five toxic effects embedded in TCMSTD; and third, the use of TCMSTD to predict toxicity based on the structural information of ingredients. A qualified database should have a reasonable data structure, a small degree of data redundancy, and accurate and reliable results included in the information and analysis system, and can track the research status of toxic-TCM in real-time to make timely integration and analysis, which is the purpose of the construction of the TCMSTD. We are confident that TCMSTD will be an effective tool for promoting TCM safety usage and providing an efficient and comprehensive platform to researchers engaged in TCM toxicity research, TCM safety regulators, and drug users for sharing information on TCM toxicology.

Compliance and ethics *The author(s) declare that they have no conflict of interest.*

Acknowledgements *This work was supported by the National Natural Science Foundation of China (81873194, 81573825, 81273998) and the National Talent Project of "Youth Qihuang Scholars" (2020). The author would like to thank Mr. Pu XUE and Mr. Moyu LIU in EHBIO Gene Technology (Beijing) Co., Ltd. for their help in the construction of the TCMSTD.*

References

Davis, A.P., Grondin, C.J., Johnson, R.J., Sciaky, D., Wiegers, J., Wiegers, T.C., and Mattingly, C.J. (2021). Comparative Toxicogenomics Database (CTD): update 2021. Nucleic Acids Res 49, D1138–D1143.

Fang, S.S., Dong, L., Liu, L., Guo, J.C., Zhao, L.H., Zhang, J.Y., Bu, D.C., Liu, X.K., Huo, P.P., Cao, W.C., et al. (2021). HERB: a high-throughput experiment- and reference-guided database of traditional Chinese medicine. Nucleic Acids Res 49, D1197–D1206.

Fang, Z., Guo, Y., Wang, Z., He, T., Wang, J., Bai, Z., and Xiao, X. (2022). Asari Radix et Rhizoma consumption lacks relevance for hepatocellular carcinoma in patients: a retrospective cohort study. Chin Herb Med 14, 470–475.

Hamosh, A., Amberger, J.S., Bocchini, C., Scott, A.F., and Rasmussen, S. A. (2021). Online Mendelian Inheritance in Man (OMIM®): Victor McKusick's magnum opus. Am J Med Genet A 185, 3259–3265.

Joo, Y.J., Newcombe, D., Nosa, V., and Bullen, C. (2020). Investigating betel nut use, antecedents and consequences: a review of literature. Subst Use Misuse 55, 1422–1442.

Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., et al. (2021). PubChem in 2021: new data content and improved web interfaces. Nucleic Acids Res 49, D1388–D1395.

Lim, E., Pon, A., Djoumbou, Y., Knox, C., Shrivastava, S., Guo, A.C., Neveu, V., and Wishart, D.S. (2010). T3DB: a comprehensively annotated database of common toxins and their targets. Nucleic Acids Res 38, D781–D786.

Liu, X., Li, H., Wang, S., Zhang, J., and Liu, D. (2021). Sesquiterpene lactones of *Aucklandia lappa*: pharmacology, pharmacokinetics, toxicity, and structure-activity relationship. Chin Herb Med 13, 167–176.

Luo, L., Yang, J., Wang, C., Wu, J., Li, Y., Zhang, X., Li, H., Zhang, H., Zhou, Y., Lu, A., et al. (2022). Natural products for infectious microbes and diseases: an overview of sources, compounds, and chemical diversities. Sci China Life Sci 65, 1123–1145.

Pence, H.E. 和 Williams, A. (2010)。ChemSpider：一个在线化学信息资源。J Chem Educ 87, 1123–1124。Piñero, J., Saüch, J., Sanz, F. 和 Furlong, L.I. (2021)。DisGeNET cytoscape 应用：探索和可视化疾病基因组数据。Comput Struct Biotechnol J 19, 2960–2967。Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y. 等 (2014)。TCMSP：用于从草药中发现药物的系统药理学数据库。J Cheminform 6, 13。Sayers, E.W., Beck, J., Bolton, E.E., Bourexis, D., Brister, J.R., Canese, K.,

Comeau, D.C., Funk, K., Kim, S., Klimke, W., 等. (2021). 国家生物技术信息中心的数据库资源. 核酸研究 49, D10–D17. UniProt, C. (2021). UniProt: 2021年的通用蛋白质知识库. 核酸研究 49, D480–D489. Xu, H.Y., Zhang, Y.Q., Liu, Z.M., Chen, T., Lv, C.Y., Tang, S.H., Zhang, X.B., Zhang, W., Li, Z.Y., Zhou, R.R., 等. (2019). ETCM: 传统中药百科全书. 核酸研究 47, D976–D982.

支持信息

支持信息可在线获取，网址为 https://doi.org/10.1007/s11427-022-2318-4。支持材料按提交原样发布，未进行排版或编辑。科学准确性和内容的责任完全由作者承担。

Pence, H.E., and Williams, A. (2010). ChemSpider: an online chemical information resource. J Chem Educ 87, 1123–1124.

Piñero, J., Saüch, J., Sanz, F., and Furlong, L.I. (2021). The DisGeNET cytoscape app: exploring and visualizing disease genomics data. Comput Struct Biotechnol J 19, 2960–2967.

Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y., et al. (2014). TCMSP: a database of systems pharmacology for drug discovery from herbal medicines. J Cheminform 6, 13.

Sayers, E.W., Beck, J., Bolton, E.E., Bourexis, D., Brister, J.R., Canese, K., Comeau, D.C., Funk, K., Kim, S., Klimke, W., et al. (2021). Database resources of the National Center for Biotechnology Information. Nucleic Acids Res 49, D10–D17.

UniProt, C. (2021). UniProt: the universal protein knowledgebase in 2021. Nucleic Acids Res 49, D480–D489.

Xu, H.Y., Zhang, Y.Q., Liu, Z.M., Chen, T., Lv, C.Y., Tang, S.H., Zhang, X.B., Zhang, W., Li, Z.Y., Zhou, R.R., et al. (2019). ETCM: an encyclopaedia of traditional Chinese medicine. Nucleic Acids Res 47, D976–D982.

SUPPORTING INFORMATION

The supporting information is available online at https://doi.org/10.1007/s11427-022-2318-4. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.